

A small cap shrinks the region until the method is gradient descent

Abstract

R-grad($\omega, \bar{\mu}$) bounds the multiplier by $\bar{\mu}$, which the asymptotic theory needs. The radius $\Delta_k = \mu_k \|g_k\|$ is then small everywhere, and a small region makes the linear term of the model dominate its quadratic term. We measure the resulting degeneration on a quartic with four known critical points, over 96 configurations. With truncated conjugate gradient the degeneration is exact and total below $\bar{\mu} = 0.1$, where every step is the Cauchy point. With the exact Moré-Sorensen solver it is asymptotic, and we fit its rate. We also reconcile two Cauchy statistics that disagree, and the reconciliation refutes the explanation we set out to confirm.

1 The claim

A cap on the multiplier holds the radius down. The model is

$$m_k(s) = f(x_k) + g_k^\top s + \frac{1}{2} s^\top H_k s, \quad (1)$$

and on a region of radius Δ the linear term is $\mathcal{O}(\Delta)$ while the quadratic term is $\mathcal{O}(\Delta^2)$. As $\Delta \rightarrow 0$ the linear term dominates and the minimiser of (1) turns towards $-g_k$. The model Hessian stops influencing the direction. The same happens to R-DFO(ω) for small ζ , which holds Δ_k below $\zeta \omega_k$.

The two subsolvers degenerate differently, and the difference is the point of this report. Truncated conjugate gradient starts from $s = 0$ with first direction $-g_k$. If the region truncates it there, the returned step is $-\Delta_k g_k / \|g_k\|$ and the cosine is one to machine precision. The exact Moré-Sorensen solver returns $s^* = -(H_k + \sigma I)^{-1} g_k$ with σ set so that the step reaches the boundary. As $\Delta \rightarrow 0$ we have $\sigma \rightarrow \infty$ and $s^* \rightarrow -\Delta_k g_k / \|g_k\|$, without ever equalling it.

2 Setup

The test function is

$$f(p) = p_1^4 - p_1^3 + \left(\frac{1}{4} - \frac{p_1}{2}\right) p_2^2 + \frac{p_2^4}{4}. \quad (2)$$

It has four critical points of three kinds. The origin is degenerate, with $\nabla^2 f$ singular and positive semidefinite, eigenvalues 0 and 1/2. The point (0.75, 0) is a non-degenerate saddle with $\lambda_{\min} = -0.25$. The two local minimisers sit at $((1 + \sqrt{5})/4, \pm \sqrt{(\sqrt{5} - 1)/4})$.

We sweep three axes. The cap $\bar{\mu}$ takes 10 values from 0.01 to 128, and ζ takes 6 values from 0.1 to 20. The subsolver is truncated conjugate gradient or exact Moré-Sorensen. The model Hessian is exact, SR1 with memory ten, or L-BFGS with memory five. Every run starts from $x_0 = (-0.5, 0.6)$ and goes through `tr_solve`.

3 Three classes of step

A single Cauchy percentage merges two different events. We separate them.

1. *Truncated at the boundary*: $\|s_k\| = \Delta_k$ with direction $-g_k$. The Hessian has touched neither the direction nor the length. This alone is gradient descent in disguise.
2. *Stopped inside after one CG iteration*: the direction is $-g_k$ and the Hessian sets the length through the exact line search on the model.
3. *Two or more CG iterations*.

Table 1 reports the three fractions. Figure 1 shows them stacked.

The degeneration is total below $\bar{\mu} = 0.1$. With the exact Hessian at $\bar{\mu} \in \{0.01, 0.05, 0.1\}$ every iteration falls in class one, and the minimum cosine over the whole run is one. The method is exactly gradient descent for two thousand iterations. Class two is almost never occupied, which says the degeneration proceeds by truncation rather than by a short interior step.

Model	$\bar{\mu}$	status	iter	trunc.	inside	2+ CG	min cos
ExactHessian	0.01	max_iter	2000	1.000	0.000	0.000	1.000000
	0.05	max_iter	2000	1.000	0.000	0.000	1.000000
	0.10	max_iter	2000	1.000	0.000	0.000	1.000000
	0.30	max_iter	2000	1.000	0.000	0.001	0.877285
	0.50	max_iter	2000	0.999	0.000	0.001	0.877285
	1.00	max_iter	2000	0.999	0.000	0.002	0.877285
	2.00	max_iter	2000	0.998	0.001	0.002	0.877285
	4.00	max_iter	2000	0.334	0.001	0.665	0.877285
	8.00	first_order	1678	0.331	0.001	0.668	0.675664
	128.00	first_order	85	0.047	0.035	0.918	0.246457
SR1	0.01	max_iter	2000	1.000	0.000	0.000	1.000000
	0.05	max_iter	2000	1.000	0.000	0.000	1.000000
	0.10	max_iter	2000	1.000	0.000	0.000	1.000000
	0.30	max_iter	2000	1.000	0.000	0.000	1.000000
	0.50	stalled	83	1.000	0.000	0.000	1.000000
	1.00	first_order	23	0.739	0.000	0.261	0.671198
	2.00	first_order	24	0.625	0.042	0.333	0.692186
	4.00	first_order	19	0.632	0.105	0.263	0.761860
	8.00	first_order	19	0.632	0.105	0.263	0.761860
	128.00	first_order	19	0.632	0.105	0.263	0.761860
LBFGS	0.01	max_iter	2000	1.000	0.000	0.000	1.000000
	0.05	max_iter	2000	1.000	0.000	0.000	1.000000
	0.10	max_iter	2000	1.000	0.000	0.000	1.000000
	0.30	max_iter	2000	1.000	0.000	0.000	1.000000
	0.50	stalled	85	1.000	0.000	0.000	1.000000
	1.00	first_order	30	0.333	0.100	0.567	0.661424
	2.00	stalled	28	0.464	0.143	0.393	0.599361
	4.00	first_order	21	0.333	0.238	0.429	0.426056
	8.00	first_order	21	0.333	0.238	0.429	0.426056
	128.00	first_order	21	0.333	0.238	0.429	0.426056

Table 1: The three classes of step under truncated conjugate gradient, as fractions of the iterations that happened. The last column is the minimum of $\cos(s_k, -g_k)$ over the run.

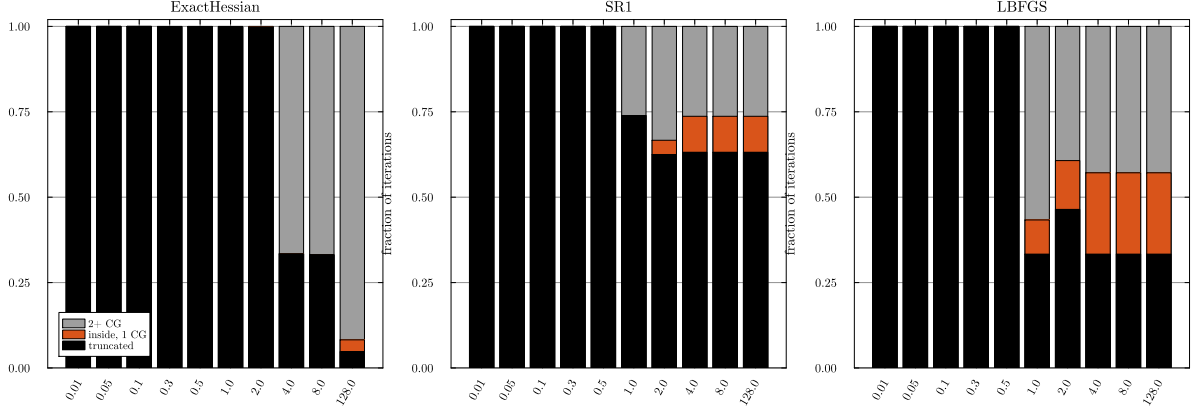


Figure 1: The three classes against $\bar{\mu}$, one panel per model Hessian, truncated conjugate gradient only.

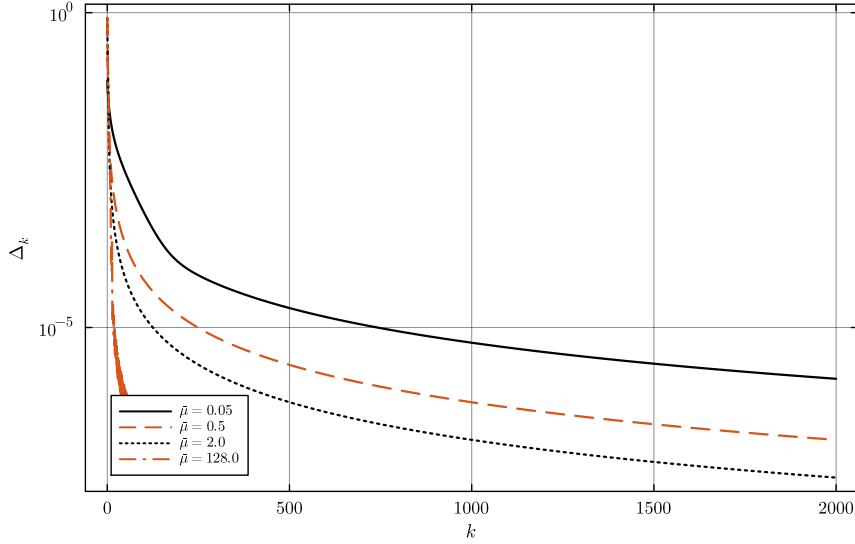


Figure 2: Δ_k against k for four caps. The cap holds the region down by orders of magnitude for the whole run.

4 Two Cauchy statistics, reconciled

The package reports `cauchy_fraction`, the fraction of iterations with $\cos(s_k, -g_k)$ within 10^{-12} of one. Counting `cg_iters == 1` tests the Krylov dimension instead. The two disagree at $\bar{\mu} = 4$, where they read 0.4885 and 0.3345, and at $\bar{\mu} = 8$, where they read 0.4815 and 0.3319. The cosine test counts more.

The natural explanation is that g_k is close to an eigenvector of H_k . The second CG direction would then add nothing to the direction while still being counted. On this function the iterates approach the axis $p_2 = 0$, where the Hessian is diagonal and the gradient is an eigenvector, so the explanation is plausible.

We tested it and it is wrong. We recorded $\sin \angle(H_k g_k, g_k)$ at every iteration, which vanishes exactly when g_k is an eigenvector of H_k , and compared the disagreeing iterations against the agreeing ones. At $\bar{\mu} = 4$ the medians are 0.707 and 0.706, which is no difference at all, and both sit near $1/\sqrt{2}$, far from an eigenvector. At $\bar{\mu} = 8$ they are 0.866 and 0.499, so the disagreeing iterations are *further* from the eigenvector case, not closer.

The mechanism is the boundary crossing. Let z_1 denote the first Steihaug iterate, the exact minimiser of (1) along $-g_k$. Truncated CG stops at the first iteration when $\|z_1\| \geq \Delta_k$ and continues when it does not. On the disagreeing iterations $\|z_1\|/\Delta_k$ has median 0.999999996 at $\bar{\mu} = 4$, so z_1 lands within a part in 10^8 of the boundary. Conjugate gradient does not truncate, takes a second iteration, and immediately backtracks to the boundary by a negligible distance. The step is on the boundary and parallel to $-g_k$ to within 10^{-12} , and `cg_iters` records two. On the agreeing iterations the same ratio has median 0.9977. All 308 disagreeing iterations exceed 0.999, against a small minority of the agreeing ones.

The two statistics answer different questions, and neither is wrong. The cosine test measures the direction that came back. The Krylov count measures the work that produced it. They separate exactly when the first Steihaug iterate lands on the boundary to within rounding, and that is a bookkeeping event rather than a property of the spectrum. Table 2 reports both alongside the two diagnostics.

Rule	param	cauchy	cg=1	gap	n	$\sin \angle$ dis.	$\sin \angle$ agr.	$\ z_1\ /\Delta$ dis.
cap	4.00	0.4885	0.3345	0.1540	308	0.707	0.706	0.999999996
cap	8.00	0.4815	0.3319	0.1496	251	0.866	0.499	0.999999989

Table 2: The two Cauchy statistics, with the near-eigenvector measure $\sin \angle(H_k g_k, g_k)$ and the boundary ratio $\|z_1\|/\Delta_k$, medians over the disagreeing and agreeing iterations. Rows with no disagreement are omitted. Exact Hessian only, since H_k is known in closed form there.

5 The angle against the radius

For truncated conjugate gradient on the truncated branch, $1 - \cos(s_k, -g_k)$ is identically zero. Not small, zero. That is the contrast this section is about.

For the exact solver we fitted the exponent rather than assuming it. Fitting $\log(1 - \cos)$ on $\log \Delta_k$ over the cap sweep gives slope 1.45 with $R^2 = 0.30$. The fit is poor and the exponent is not two.

The reason is that Δ_k is not the parameter the expansion is in. Write $s^* = -(H + \sigma I)^{-1}g$ in the eigenbasis of H and expand for large σ , using $\sigma \sim \|g\|/\Delta$. This gives

$$1 - \cos(s^*, -g) \sim \frac{1}{2} \text{Var}_w(\lambda) \mu^2, \quad \mu = \frac{\Delta}{\|g\|}, \quad (3)$$

where w_i are the weights of g on the eigenvectors of H and Var_w is the weighted variance of the spectrum. The expansion parameter is μ , not Δ .

That distinction is exactly what a cap enforces. `R-grad`($\omega, \bar{\mu}$) sets $\Delta_k = \mu_k \|g_k\|$ and pins μ_k at $\bar{\mu}$, so μ_k is constant along a run. The radius goes to zero only because $\|g_k\|$ does. No exponent in μ can be fitted from one run, and the variation has to come from pooling runs at different caps.

Fitting (3) that way, over 3362 iterations with $\mu \leq 0.5$, gives slope 1.961 with $R^2 = 0.9990$ and intercept -0.377 against the predicted $\log_{10}(1/2) = -0.301$. The second order is confirmed. Figure 3 shows both fits.

The residual decrease of $1 - \cos$ along a run therefore comes from $\text{Var}_w(\lambda)$ and not from Δ_k . The iterates approach the degenerate origin along a direction on which the Hessian is nearly singular, the gradient aligns with an eigenvector, and the weighted variance collapses. The near-eigenvector effect is real. It drives the exact solver's angle to zero. It is not what separates the two Cauchy statistics of Section 4.

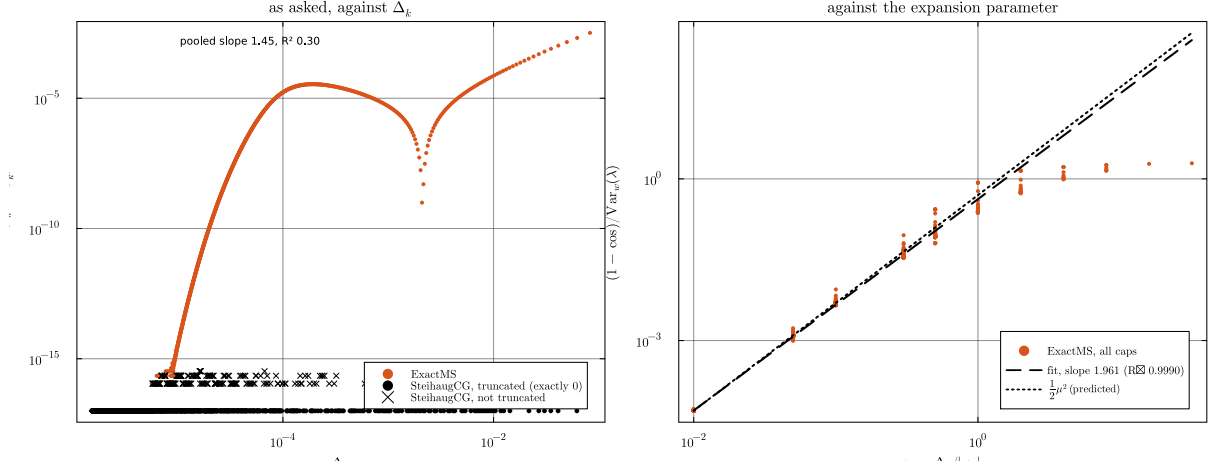


Figure 3: Left, $1 - \cos(s_k, -g_k)$ against Δ_k as asked, with the truncated Steihaug iterations at exactly zero placed on the floor of the axis. Right, the same quantity divided by $\text{Var}_w(\lambda)$ against μ_k , pooled over the cap sweep, with the fitted line and the predicted $\frac{1}{2}\mu^2$.

6 Limit points

A category column invites the reading that a trapped run converged somewhere else. We report the distance to the nearest critical point and $\|g_k\|$ at the last iterate, with a work measure, so that the row survives a change of budget. Table 3 gives them. No configuration is dropped. The trapped ones are the evidence.

Subsolver	Model	$\bar{\mu}$	status	iter	dist	$\ g\ $	∇f evals
SteihaugCG	ExactHessian	0.01	max_iter	2000	1.421e-02	6.172e-04	2001
SteihaugCG	ExactHessian	0.05	max_iter	2000	3.191e-03	3.068e-05	2001
SteihaugCG	ExactHessian	0.10	max_iter	2000	1.624e-03	7.927e-06	2001
SteihaugCG	ExactHessian	0.30	max_iter	2000	5.500e-04	9.082e-07	2001
SteihaugCG	ExactHessian	0.50	max_iter	2000	3.310e-04	3.289e-07	2001
SteihaugCG	ExactHessian	1.00	max_iter	2000	1.658e-04	8.253e-08	2001
SteihaugCG	ExactHessian	2.00	max_iter	2000	8.303e-05	2.068e-08	2001
SteihaugCG	ExactHessian	4.00	max_iter	2000	4.142e-05	7.280e-09	2001
SteihaugCG	ExactHessian	8.00	first_order	1678	1.698e-05	9.992e-10	1679
SteihaugCG	ExactHessian	128.00	first_order	85	1.754e-05	9.985e-10	86
ExactMS	ExactHessian	0.01	max_iter	2000	1.423e-02	6.195e-04	2001
ExactMS	ExactHessian	0.05	max_iter	2000	3.199e-03	3.083e-05	2001
ExactMS	ExactHessian	0.10	max_iter	2000	1.629e-03	7.976e-06	2001
ExactMS	ExactHessian	0.30	max_iter	2000	5.503e-04	9.093e-07	2001
ExactMS	ExactHessian	0.50	max_iter	2000	3.310e-04	3.289e-07	2001
ExactMS	ExactHessian	1.00	max_iter	2000	1.658e-04	8.253e-08	2001
ExactMS	ExactHessian	2.00	max_iter	2000	8.303e-05	2.068e-08	2001
ExactMS	ExactHessian	4.00	max_iter	2000	4.155e-05	5.180e-09	2001
ExactMS	ExactHessian	8.00	max_iter	2000	2.079e-05	1.297e-09	2001
ExactMS	ExactHessian	128.00	first_order	146	1.819e-05	9.924e-10	147

Table 3: Limit points as distance rather than as a category, exact Hessian. A run at **max_iter** has not converged anywhere, and its distance says how far from a critical point it stopped.

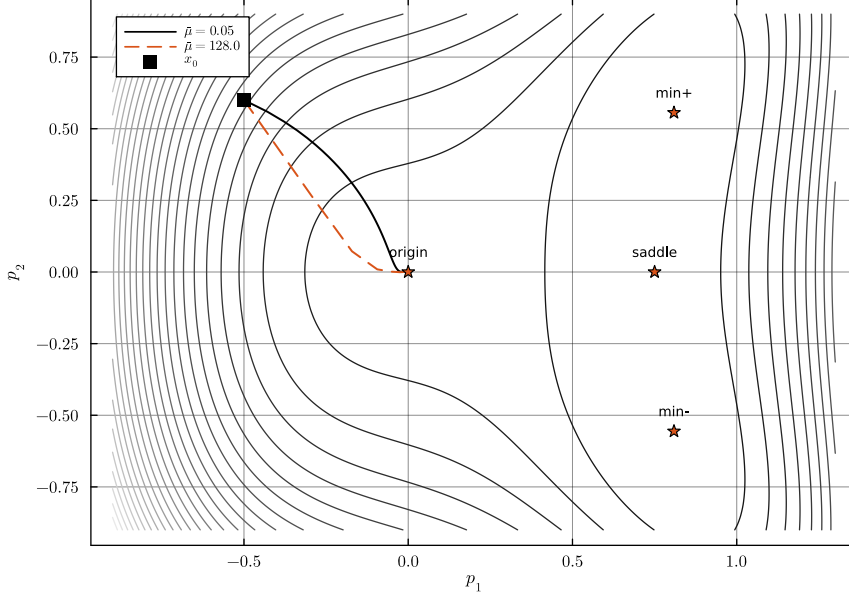


Figure 4: Iterate paths for one small and one large cap over the contours of f , with the four critical points marked.

7 Scope

With the exact Hessian these runs end at the degenerate origin, where $\nabla^2 f$ is singular. Part II's inactivity theory assumes a positive definite Hessian at the limit. An activity statistic collected on such a run is therefore not evidence about inactivity. The constraint stays active there for a reason that has nothing to do with the cap. We report the activity column in Table 3 and draw no inactivity reading from it.

Which critical point a run reaches, and why the exact Hessian is the model that lands on the degenerate one, is a separate question. We record what happens and do not investigate it here.

8 Where this belongs

This is a candidate for the section of Part III on what the rules do, labelled **sec: designed** there, as the subsection that gives the cost of the cap. It sits naturally after the radius asymptotics and before the threshold material, because it establishes what a small region does to the step before any threshold in a curvature quantity is estimated. The reconciliation of Section 4 belongs with the diagnostics rather than in the main line, since it is about two package statistics rather than about a radius rule.